

Dyadic Primitive Ladders and Mellin–Newton Antiderivatives

for the Fabius–Rvachev System

Polynomial closed forms, fractional and negative powers, transformed bumps, higher-order operators, inverse quantiles, and frontier questions

Frontier research report based on the recursive `*.tex` corpus under

`Analysis/FabiusFunction/docs`

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Abstract

Let F be the bounded Fabius distribution function, let up be Rvachev's compactly supported even bump, let $\mathcal{F}$ be the signed global Fabius extension, and let $Q = F^{-1}$. The existing PROVEIT corpus develops their differential equations, exact dyadic arithmetic, moment recurrences, half-base q -binomial formulae, infinite sinc products, lower-Lambert endpoint asymptotics, and Bell–Bernoulli coefficient calculus, but does not contain a systematic calculus for antiderivatives of monomially weighted Fabius functions. This report supplies such a calculus.

The key observation is an explicit primitive ladder. If

$$J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m),$$

then $J_0 = \mathcal{F}$ and $J'_{m+1} = J_m$. Consequently every polynomial weight closes after finitely many integrations by parts. For $p \in \mathbb{N}$, $n \geq 1$, and $0 \leq x \leq 1$ we prove

$$I_0^n[x^p F(x)] = \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k}),$$

where I_0^n is the normalized n -fold primitive and $n^{\bar{k}}$ is a rising factorial. The ordinary antiderivative is the case $n = 1$. A piecewise formula extends the result to the globally bounded F .

For an arbitrary complex exponent α , the finite formula becomes a normally convergent Mellin–Newton series

$$I_0^n[x^\alpha F(x)] = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{\alpha-k} F(x/2^{n+k}).$$

It is entire in α , terminates exactly at nonnegative integers, and becomes a positive-term series at negative integers. Complete integrals are governed by the entire moment interpolation $D(s) = \mathbb{E}X^s$:

$$\int_0^1 x^\alpha F(x) dx = \frac{1 - D(\alpha + 1)}{\alpha + 1} = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{d_{k+1}}{k + 1}.$$

This identifies dyadic Fabius samples with forward differences in the exponent, proves strict complete monotonicity and total positivity on the real exponent axis, and connects the antiderivative problem directly to the sinc-product cumulants and Bell–Bernoulli moments.

The same ladder yields shifted closed forms for $x^p \text{up}(x)$ on $[-1, 1]$ and formulas for the globally identical bounded-CDF transforms $x^p F(1 - x) = x^p(1 - F(x))$, including their meromorphic exponent continuations. For the inverse function we obtain the compact quantile identity

$$\int_0^y Q(v)^\alpha dv = yQ(y)^\alpha - \alpha \int_0^{Q(y)} x^{\alpha-1} F(x) dx.$$

In particular,

$$\int Q(y) dy = yQ(y) - F(Q(y)/2) + C.$$

Higher inverse primitives lead naturally to powers F^j , or equivalently to order statistics of independent Fabius random variables.

Exact reciprocal-dyadic formulae, Thue–Morse and q -binomial substitutions, numerical experiments, a negative-power tail model driven by the proved log-periodic endpoint asymptotic, conjectures, and a Lean formalization plan complete the report. All statements are labeled as inherited, proved here, numerical, or conjectural. Priority outside the pinned corpus is not asserted.

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1 Audit boundary, novelty discipline, and claim status

1.1 Pinned corpus and reading method

The source boundary is the exact ProveIt commit printed on the title page [12]. Its recursive documentation tree contains 76 indexed `.tex` paths. Every path was inventoried and searched; the living expositions, source-paper transcriptions, archived revisions, and frontier drafts were then grouped into mathematical lineages so that superseded copies were not mistaken for independent results. The families most relevant to the present problem—the primary Fabius–Rvachev exposition, repeated-integration studies, inverse-function reports, moment and q -calculus chapters, endpoint asymptotics, dyadic formulae, self-sampling/Appell material, and spectral/frontier dossiers—were read in depth. The full path inventory is recorded in [section C](#).

Targeted searches were made for *antiderivative*, *primitive*, *monomial*, products of F with powers, inverse-function areas, and candidate formula fragments such as $F(x/2)$. The archive does contain important repeated-integration work, but there the iteration is applied to boxes, splines, refinements, or convolution approximants in order to construct up; it is not a calculus of $\int x^p F(x) dx$. The inverse reports concern endpoint inversion and Lambert coordinates, not quantile antiderivatives. The Newton-basis frontier report concerns exponent sequences in infinite sinc products, not Newton interpolation in the power α of the integrand.

Novelty scope. The formula package below was not found in the pinned repository after path, phrase, and formula searches. This is a repository-relative novelty statement, not a claim of first publication in the worldwide literature. Classical ingredients such as repeated integration by parts, Newton series, Mellin transforms, order statistics, and total positivity are of course standard; the point is their unusually exact closure in the Fabius–Rvachev system.

1.2 Status convention

The report uses four statuses.

Inherited	A theorem or definition already present in the pinned corpus, usually with a matching Lean declaration.
Proved here	A complete paper proof is supplied below from inherited facts. It is not yet claimed to be formalized.
Numerical	A reproducible finite-precision or exact-rational experiment in the accompanying Python program.
Conjectural	A statement suggested by the proved structure and experiments, with the missing analytic step identified.

1.3 Main new theorem package

The repository search found no counterpart of the following combined package:

1. the dyadic primitive ladder $J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m)$;
2. the terminating polynomial formula for every monomial and every order of integration;
3. the normally convergent complex-exponent Mellin–Newton continuation;
4. the recovery of reciprocal-dyadic samples as forward differences in the exponent;
5. complete-moment generating functions and strict total positivity of the exponent kernel;

6. shifted polynomial primitives of up and a careful separation of $F(1 - x)$ from the globally compact bump;
7. the inverse-quantile primitive and its higher-order order-statistic hierarchy;
8. exact dyadic formulae for weighted primitives and inverse areas;
9. the negative-power tail model obtained by combining Newton terms with the proved log-periodic endpoint asymptotic.

2 Inherited Fabius–Rvachev data and notation

2.1 Three functions and one random variable

We keep the repository’s three-function convention. Historically, the distribution goes back to Jessen–Wintner and Fabius, while the compactly supported form was developed by the Rvachevs [1, 2, 3, 4]. The arithmetic and dyadic normalizations used here follow the modern treatments of Arias de Reyna and Haugland [5, 6, 7].

Definition 2.1. The *bounded Fabius function* $F : \mathbb{R} \rightarrow [0, 1]$ is zero on $(-\infty, 0]$, one on $[1, \infty)$, and on the unit interval satisfies

$$F(1 - x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2). \quad (2.1)$$

The *Rvachev up-function* is

$$\text{up}(x) = F(1 - |x|), \quad \text{supp up} = [-1, 1]. \quad (2.2)$$

The *signed global extension* $\mathcal{F}$ agrees with F on $[0, 1]$, vanishes on the negative half-line, and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (2.3)$$

On $[-1, 1]$ one has $\text{up}(x) = \mathcal{F}(x + 1)$.

The bounded function is the distribution function of

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]. \quad (2.4)$$

The law is symmetric: $X \stackrel{d}{=} 1 - X$. Its moments are

$$d_s := D(s) := \mathbb{E}X^s \quad (s \in \mathbb{C}), \quad d_n = D(n) \quad (n \in \mathbb{N}). \quad (2.5)$$

The notation d_s is used only informally for noninteger s ; D is the analytic object. The corpus proves for integer $n \geq 1$

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) \, dt, \quad (2.6)$$

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k, \quad (2.7)$$

$$F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (2.8)$$

Infinite flatness at both endpoints implies that $D(s) = \mathbb{E}X^s$ extends to an entire function: for every compact $K \subset \mathbb{C}$, the integrand X^s is uniformly integrable for $s \in K$. It also implies

$$d_n = \mathcal{O}_A(n^{-A}) \quad \text{for every } A > 0. \quad (2.9)$$

This last, deliberately weak form is enough for all Newton-series convergence arguments.

2.2 Integral operators and factorials

For $n \geq 1$ define the normalized n -fold primitive from the origin by

$$(I_0^n f)(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) \, dt, \quad I_0^0 f = f. \quad (2.10)$$

Thus $(I_0^n f)^{(n)} = f$, and all derivatives of orders $0, \dots, n-1$ vanish at zero. For a base point a we similarly write

$$(I_a^n f)(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) \, dt. \quad (2.11)$$

We use

$$z^{\underline{k}} = z(z-1) \cdots (z-k+1), \quad z^{\overline{k}} = z(z+1) \cdots (z+k-1), \quad (2.12)$$

with either product equal to 1 at $k = 0$. Hence $\binom{z}{k} = z^{\underline{k}}/k!$ and $n^{\overline{k}} = (n+k-1)!/(n-1)!$ for $n \in \mathbb{N}_{>0}$.

3 The dyadic primitive ladder

3.1 Exact closure under integration

Theorem 3.1 (Dyadic primitive ladder). *For $m \in \mathbb{N}$ put*

$$J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m). \quad (3.1)$$

Then

$$J_0 = \mathcal{F}, \quad J'_{m+1} = J_m, \quad J_m^{(m)} = \mathcal{F}. \quad (3.2)$$

For $0 \leq x \leq 1$,

$$(I_0^m \mathcal{F})(x) = J_m(x) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m). \quad (3.3)$$

Proof. Differentiate (3.1) and use (2.3):

$$J'_{m+1}(x) = 2^{\binom{m+1}{2}} 2^{-(m+1)} 2\mathcal{F}(x/2^m) = 2^{\binom{m}{2}} \mathcal{F}(x/2^m) = J_m(x).$$

Every J_m and every derivative below order m vanishes at zero because $\mathcal{F}$ is flat there. Hence J_m is exactly the normalized primitive (2.10). On the unit interval $\mathcal{F} = F$ and all arguments $x/2^m$ remain in $[0, 1]$. $\square$

Interpretation. The usual obstruction to a finite antiderivative formula is that integrating a special function creates a new special function. Here integration creates no new object: it only dilates the argument by 2^{-1} and multiplies by a triangular power of two. Polynomial weights can therefore be removed by finitely many integrations by parts.

3.2 A useful moment bound for the ladder

Lemma 3.2. *For $m \geq 1$ and $0 \leq x \leq 1$,*

$$0 \leq J_m(x) \leq x^m J_m(1) = x^m \frac{d_m}{m!}. \quad (3.4)$$

Proof. The Cauchy representation and monotonicity of F give

$$\begin{aligned} J_m(x) &= \frac{x^m}{(m-1)!} \int_0^1 (1-u)^{m-1} F(xu) \, du \\ &\leq \frac{x^m}{(m-1)!} \int_0^1 (1-u)^{m-1} F(u) \, du = x^m J_m(1). \end{aligned}$$

If X has distribution function F , Tonelli and symmetry yield

$$J_m(1) = \frac{1}{m!} \mathbb{E}(1-X)^m = \frac{d_m}{m!}.$$

□

This estimate is the convergence engine for the complex-exponent theory. Notice the exact cancellation that will later occur:

$$\frac{n^{\bar{k}}}{(n+k)!} = \frac{1}{(n-1)!(n+k)}.$$
 (3.5)

4 Polynomial weights: terminating closed forms

4.1 A polynomial theorem

Theorem 4.1 (Polynomially weighted n -fold primitive). *Let P be a polynomial of degree d and $n \geq 1$. On every interval on which the signed extension is used consistently,*

$$I_0^n[P\mathcal{F}](x) = \sum_{k=0}^d (-1)^k \frac{n^{\bar{k}}}{k!} P^{(k)}(x) 2^{\binom{n+k}{2}} \mathcal{F}(x/2^{n+k}).$$
 (4.1)

For $0 \leq x \leq 1$, $\mathcal{F}$ may be replaced by the bounded F throughout.

Proof. Write $J_m = I_0^m \mathcal{F}$. Repeated integration by parts gives

$$I_0^n[P\mathcal{F}] = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} P^{(k)} J_{n+k}.$$

The coefficient can be proved by induction on k : each new integration transfers a derivative to P and multiplies the Cauchy-kernel order by $(n+k)/(k+1)$. Since $P^{(k)} = 0$ for $k > d$, the expansion terminates. Substitution of (3.1) proves the formula. One can also verify it directly by differentiating n times: adjacent terms cancel telescopically, leaving $P\mathcal{F}$. □

4.2 Monomial formula and examples

Corollary 4.2 (Closed form for $p \in \mathbb{N}$). *For $p \in \mathbb{N}$, $n \geq 1$, and $0 \leq x \leq 1$,*

$$I_0^n[x^p F(x)] = \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k}).$$
 (4.2)

In particular, an ordinary antiderivative is

$$\int x^p F(x) \, dx = \sum_{k=0}^p (-1)^k p^{\bar{k}} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C.$$
 (4.3)

The first cases are strikingly compact:

$$\int F(x) \, dx = F(x/2) + C, \quad (4.4)$$

$$\int xF(x) \, dx = xF(x/2) - 2F(x/4) + C, \quad (4.5)$$

$$\int x^2 F(x) \, dx = x^2 F(x/2) - 4xF(x/4) + 16F(x/8) + C, \quad (4.6)$$

$$\int x^3 F(x) \, dx = x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 512F(x/16) + C. \quad (4.7)$$

The powers $1, 2, 16, 512, \dots$ are $2^{\binom{k+1}{2}}$, not a geometric sequence; the triangular exponent is forced by iterating the dilation derivative.

Remark 4.3 (General n th antiderivative). Formula (4.2) is the unique n th antiderivative whose first n Taylor coefficients at zero vanish. The general n th antiderivative is obtained by adding an arbitrary polynomial of degree at most $n - 1$.

4.3 A recurrence in the exponent

Let

$$A_\alpha(x) = \int_0^x t^\alpha F(t) \, dt. \quad (4.8)$$

For nonnegative integer α this is already covered by (4.3), but the recurrence below remains meaningful for every complex exponent.

Proposition 4.4 (Dilation recurrence). *For $0 < x \leq 1$ and $\alpha \in \mathbb{C}$,*

$$A_\alpha(x) = x^\alpha F(x/2) - \alpha 2^\alpha A_{\alpha-1}(x/2). \quad (4.9)$$

For $0 < x \leq 1/2$ one also has

$$(\alpha + 1)A_\alpha(x) = x^{\alpha+1} F(x) - 2^{-\alpha-1} A_{\alpha+1}(2x). \quad (4.10)$$

Proof. For the first formula integrate by parts against the primitive $F(t/2)$ and substitute $t = 2u$ in the remaining integral. For the second, integrate A_α by parts in the opposite direction and use $F'(t) = 2F(2t)$ on the left half of the unit interval. Infinite flatness makes the endpoint terms at zero vanish for every complex α . $\square$

4.4 The globally bounded F : exact piecewise formula

The signed extension is the natural object for the pure ladder, but applications often want the bounded CDF. The normalized primitive then has a simple piecewise completion.

Theorem 4.5 (Global bounded-CDF formula). *Let $p \in \mathbb{N}$ and $n \geq 1$. For $x \leq 0$,*

$$I_0^n[t^p F(t)](x) = 0. \quad (4.11)$$

For $0 \leq x \leq 1$ use (4.2). For $x \geq 1$,

$$I_0^n[t^p F(t)](x) = \frac{p!}{(p+n)!} x^{p+n} - \sum_{\ell=0}^{n-1} \frac{(-1)^\ell x^{n-1-\ell}}{\ell!(n-1-\ell)!} \frac{d_{p+\ell+1}}{p+\ell+1}. \quad (4.12)$$

Proof. For $x \geq 1$, write $F = 1 - (1 - F)$. The primitive of t^p is the first term. The survival function $1 - F$ is supported on $[0, 1]$, so expand $(x - t)^{n-1}$ and use

$$\int_0^1 t^m (1 - F(t)) dt = \frac{\mathbb{E} X^{m+1}}{m+1} = \frac{d_{m+1}}{m+1}.$$

The left half-line is immediate from the bounded extension $F = 0$ there. $\square$

Thus outside $[0, 1]$ the answer is a polynomial whose coefficients are exact rational moments. The apparently non-elementary content is confined to the active unit interval.

5 Fractional, negative, and complex powers

5.1 The Mellin–Newton expansion

Fix the real logarithm for $x > 0$, so $x^\alpha = \exp(\alpha \log x)$.

Theorem 5.1 (Complex-exponent n -fold primitive). *For every $n \geq 1$, $0 < x \leq 1$, and $\alpha \in \mathbb{C}$,*

$$I_0^n[t^\alpha F(t)](x) = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{\alpha-k} F(x/2^{n+k}). \quad (5.1)$$

The series converges absolutely and locally uniformly in α and uniformly for x in each compact subinterval of $(0, 1]$. It therefore defines an entire function of α . For $\alpha = p \in \mathbb{N}$ it terminates and becomes (4.2).

Proof. In the Cauchy kernel (2.10), write

$$t^\alpha = x^\alpha \left(1 - \frac{x-t}{x}\right)^\alpha = x^\alpha \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{(x-t)^k}{x^k}.$$

On $[\epsilon, x]$ the binomial series is uniformly convergent. Its k th integrated term is

$$(-1)^k \binom{\alpha}{k} x^{\alpha-k} n^{\bar{k}} J_{n+k}(x),$$

which is the term in (5.1). To pass to zero and prove normal convergence, apply theorem 3.2 and (3.5):

$$\left| \binom{\alpha}{k} n^{\bar{k}} x^{\alpha-k} J_{n+k}(x) \right| \leq \frac{x^{\Re \alpha + n}}{(n-1)!} \left| \binom{\alpha}{k} \right| \frac{d_{n+k}}{n+k}. \quad (5.2)$$

On a compact set of exponents, $|\binom{\alpha}{k}|$ has at most polynomial growth in k , whereas (2.9) beats every polynomial. The Weierstrass test proves all stated forms of convergence and justifies termwise integration. $\square$

For $n = 1$ this reads

$$A_\alpha(x) = \sum_{k=0}^{\infty} (-1)^k \alpha^{\bar{k}} 2^{\binom{k+1}{2}} x^{\alpha-k} F(x/2^{k+1}). \quad (5.3)$$

The infinite series is not merely an asymptotic integration-by-parts expansion: endpoint flatness makes it globally convergent for every complex exponent.

5.2 Negative powers and positivity

For $m \in \mathbb{N}_{>0}$, $(-1)^k - m^{\bar{k}} = m^{\bar{k}}$, so

$$A_{-m}(x) = \sum_{k=0}^{\infty} m^{\bar{k}} 2^{\binom{k+1}{2}} x^{-m-k} F(x/2^{k+1}). \quad (5.4)$$

Every term is positive. Therefore every truncation is a rigorous lower bound, and the remainder has no cancellation. This is unusually favorable for certified computation. For a general negative real exponent $\alpha = -a$, $a > 0$,

$$A_{-a}(x) = x^{-a} \sum_{k \geq 0} \binom{a+k-1}{k} k! 2^{\binom{k+1}{2}} x^{-k} F(x/2^{k+1}). \quad (5.5)$$

At $x = 1$, the factorial and dyadic factors collapse to moments, as developed in [section 6](#).

Remark 5.2 (Why all negative powers are integrable). Although x^α itself is nonintegrable at zero when $\Re \alpha \leq -1$, the Fabius factor is flatter than every power. For each N there is C_N with $F(x) \leq C_N x^N$ near zero, so $x^\alpha F(x)$ is integrable for every fixed complex α . The entire dependence on α is a direct analytic expression of infinite flatness.

5.3 Exponent interpolation and exact forward differences

Define

$$B_{n,x}(\alpha) = x^{-\alpha} I_0^n [t^\alpha F(t)](x). \quad (5.6)$$

Then (5.1) is a Newton series in α .

Theorem 5.3 (Exponent–dyadic duality). For $k \in \mathbb{N}$,

$$\Delta^k B_{n,x}(0) = (-1)^k n^{\bar{k}} 2^{\binom{n+k}{2}} x^{-k} F(x/2^{n+k}), \quad (5.7)$$

where $\Delta h(\alpha) = h(\alpha + 1) - h(\alpha)$. Equivalently,

$$F(x/2^{n+k}) = \frac{(-1)^k x^k}{n^{\bar{k}} 2^{\binom{n+k}{2}}} \Delta^k B_{n,x}(0). \quad (5.8)$$

At $n = x = 1$,

$$d_{k+1} = (-1)^k (k+1) \Delta^k M(0), \quad M(\alpha) := A_\alpha(1). \quad (5.9)$$

Proof. The Newton basis satisfies $\Delta^j \binom{\alpha}{k} |_{\alpha=0} = \delta_{jk}$. Apply this to (5.1) after multiplication by $x^{-\alpha}$. The last identity uses (2.8). $\square$

A new reconstruction principle. Knowing the complete integral $\alpha \mapsto \int_0^1 x^\alpha F(x) dx$ at the nonnegative integers determines every reciprocal-dyadic sample by finite differences. Conversely, the reciprocal-dyadic ladder is precisely the Newton coefficient sequence of the exponent interpolation. Thus dyadic argument scale and discrete exponent scale are dual coordinates.

6 Complete integrals, moments, and special-function structure

6.1 Entire Mellin interpolation

Theorem 6.1 (Complete weighted integral). *The function*

$$M(\alpha) = \int_0^1 x^\alpha F(x) \, dx \quad (6.1)$$

is entire and satisfies

$$M(\alpha) = \frac{1 - D(\alpha + 1)}{\alpha + 1} = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} \frac{d_{k+1}}{k+1}. \quad (6.2)$$

The quotient has a removable singularity at $\alpha = -1$, with

$$M(-1) = -D'(0) = -\mathbb{E} \log X = \sum_{k \geq 0} \frac{d_{k+1}}{k+1}. \quad (6.3)$$

For $p \in \mathbb{N}$,

$$\int_0^1 x^p F(x) \, dx = \frac{1 - d_{p+1}}{p+1}. \quad (6.4)$$

For $m \in \mathbb{N}_{>0}$,

$$M(-m) = \sum_{k \geq 0} \binom{m+k-1}{k} \frac{d_{k+1}}{k+1}. \quad (6.5)$$

Proof. For $\Re \alpha > -1$, Stieltjes integration by parts gives

$$D(\alpha + 1) = \int_0^1 x^{\alpha+1} \, dF(x) = 1 - (\alpha + 1)M(\alpha).$$

Both sides extend entirely because of endpoint flatness, proving the first equality and the removable value. The second equality is (5.1) at $n = x = 1$, followed by $2^{\binom{k+1}{2}} F(2^{-k-1}) = d_{k+1}/(k+1)!$ and $\alpha^{\underline{k}} = k! \binom{\alpha}{k}$. The integer and negative-integer cases follow. $\square$

The complete n -fold primitive has an equally clean series.

Corollary 6.2. *For $n \geq 1$ and $\alpha \in \mathbb{C}$,*

$$I_0^n[t^\alpha F(t)](1) = \frac{1}{(n-1)!} \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{d_{n+k}}{n+k}. \quad (6.6)$$

6.2 Ordinary and exponential generating functions

Let

$$G(z) = \mathbb{E} e^{zX} = \sum_{n \geq 0} d_n \frac{z^n}{n!}. \quad (6.7)$$

The corpus proves $G(2z) = ((e^z - 1)/z)G(z)$. The complete polynomial integrals have the exponential generating function

$$\sum_{p \geq 0} M(p) \frac{z^p}{p!} = \int_0^1 e^{zx} F(x) \, dx = \frac{e^z - G(z)}{z}, \quad (6.8)$$

with the removable value $1/2$ at zero. Their ordinary generating function is

$$\sum_{p \geq 0} M(p) z^{p+1} = -\log(1-z) - \sum_{n \geq 1} \frac{d_n z^n}{n} = \mathbb{E} \log \frac{1-zX}{1-z}, \quad |z| < 1. \quad (6.9)$$

For the positive half of the up-function,

$$\sum_{p \geq 0} z^{p+1} \int_0^1 x^p \text{up}(x) \, dx = \mathbb{E}[-\log(1-zX)]. \quad (6.10)$$

These identities turn moment recurrences into recurrences for whole families of weighted antiderivatives.

6.3 The sinc product, Bernoulli cumulants, and Bell moments

Put $Y = 2X - 1$. Its density is up on $[-1, 1]$, and its moment generating function is the hyperbolic continuation of the infinite sinc product:

$$C(z) = \mathbb{E} e^{zY} = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (6.11)$$

The even cumulants are

$$\kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (6.12)$$

With the corpus notation $c_r = \mathbb{E} Y^{2r} = \int_{-1}^1 x^{2r} \text{up}(x) \, dx$, the complete exponential Bell polynomials give

$$c_r = \mathcal{B}_{2r}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2r}). \quad (6.13)$$

Consequently the entire moment interpolation has the absolutely convergent Bell–Bernoulli form

$$D(s) = 2^{-s} \sum_{r=0}^{\infty} \binom{s}{2r} c_r = 2^{-s} \sum_{r=0}^{\infty} \binom{s}{2r} \mathcal{B}_{2r}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2r}). \quad (6.14)$$

Combining (6.14) with (6.2) expresses every fractional or negative complete Fabius integral in Bernoulli numbers and Bell polynomials. This is not necessarily the fastest numerical representation, but it makes the connection with the corpus's all-orders coefficient calculus explicit.

6.4 Complete monotonicity and strict total positivity

Theorem 6.3 (Exponent kernel). *For real α and every $m \in \mathbb{N}$,*

$$(-1)^m M^{(m)}(\alpha) = \int_0^1 x^\alpha (-\log x)^m F(x) \, dx > 0. \quad (6.15)$$

Thus M is strictly completely monotone on the whole real axis and strictly log-convex. More strongly, if $a_1 < \dots < a_r$ and $b_1 < \dots < b_r$ are real, then

$$\det[M(a_i + b_j)]_{i,j=1}^r > 0. \quad (6.16)$$

In particular, every Hankel determinant $\det[M(\alpha + i + j)]_{i,j=0}^{r-1}$ is positive.

Proof. Differentiation under the integral is justified by flatness, and the integrand in (6.15) is positive on a set of positive measure. Log-convexity follows from Cauchy–Schwarz. For total positivity, apply Andreief's identity to the positive measure $F(x) \, dx$:

$$\det[M(a_i + b_j)] = \frac{1}{r!} \int_{(0,1)^r} \det[x_j^{a_i}] \det[x_j^{b_i}] \prod_{j=1}^r F(x_j) \, dx_j.$$

On the chamber $0 < x_1 < \dots < x_r < 1$, both generalized Vandermonde determinants have the same strict sign; their product is positive. The measure of that chamber is positive. $\square$

Remark 6.4. The theorem supplies a large family of nonlinear inequalities among the exact values $M(p) = (1 - d_{p+1})/(p+1)$. For example,

$$M(a+b)M(c+d) > M(a+d)M(c+b)$$

whenever $a < c$ and $b < d$. Such inequalities are invisible in the triangular moment recurrence but automatic in the exponent-integral representation.

7 Reflection, survival, and Rvachev's up-function

7.1 The two reflected bounded-CDF transforms, and the genuinely different bump

For the bounded CDF with its constant extensions, symmetry is global:

$$F(1-x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (7.1)$$

Indeed, the two sides agree by the usual symmetry on $[0, 1]$, are both 1 for $x \leq 0$, and are both 0 for $x \geq 1$. Thus the two transformations explicitly requested in the problem have the same antiderivative theory for the bounded F .

The compact bump is different on the negative half-line:

$$\text{up}(x) = F(1 - |x|), \quad \text{up}(x) = F(1 - x) = 1 - F(x) \quad (x \geq 0). \quad (7.2)$$

It vanishes outside $[-1, 1]$, whereas the reflected/survival transform equals 1 on $(-\infty, 0]$. If F is replaced by the signed global extension $\mathcal{F}$, then $\mathcal{F}(1-x)$ and $1 - \mathcal{F}(x)$ are genuinely different outside the unit interval. Domain labels are therefore still essential.

7.2 The survival transform $1 - F(x)$

Theorem 7.1 (Weighted survival primitives). *For $p \in \mathbb{N}$, $n \geq 1$, and $0 \leq x \leq 1$,*

$$I_0^n[t^p F(1-t)](x) = I_0^n[t^p(1-F(t))](x) = \frac{p!}{(p+n)!} x^{p+n} - I_0^n[t^p F(t)](x), \quad (7.3)$$

where the last term is the finite sum (4.2). For complex α with $\Re \alpha > -1$,

$$I_0^n[t^\alpha F(1-t)](x) = I_0^n[t^\alpha(1-F(t))](x) = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha+n+1)} x^{\alpha+n} - I_0^n[t^\alpha F(t)](x). \quad (7.4)$$

The right side meromorphically continues to all α , with possible simple poles at $-1, -2, \dots, -n$.

For $x \geq 1$, the survival primitive becomes a polynomial of degree $n-1$:

$$I_0^n[t^p(1-F(t))](x) = \sum_{\ell=0}^{n-1} \frac{(-1)^\ell x^{n-1-\ell}}{\ell!(n-1-\ell)!} \frac{d_{p+\ell+1}}{p+\ell+1}. \quad (7.5)$$

For integer p and $x \leq 0$, since $1 - F(t) = 1$ along the integration path,

$$I_0^n[t^p(1 - F(t))](x) = \frac{p!}{(p+n)!} x^{p+n}. \quad (7.6)$$

At $x = 1$ the positive half of up has the Mellin transform

$$U(\alpha) := \int_0^1 x^\alpha \text{up}(x) \, dx = \frac{D(\alpha+1)}{\alpha+1}. \quad (7.7)$$

Its meromorphic continuation has exactly one pole, at $\alpha = -1$, with residue 1. The finite part is

$$\text{FP}_{\alpha=-1} U(\alpha) = D'(0) = \int_0^1 \frac{\text{up}(x) - 1}{x} \, dx. \quad (7.8)$$

This one-pole simplification is a consequence of the infinite flatness of $\text{up}(x) - 1$ at zero.

7.3 A shifted primitive ladder for the compact bump

Theorem 7.2 (Shifted Rvachev ladder). *For $m \geq 1$ and $-1 \leq x \leq 1$, define*

$$K_m(x) = 2^{\binom{m}{2}} F((x+1)/2^m). \quad (7.9)$$

Then $K'_1 = \text{up}$, $K'_{m+1} = K_m$, and K_m is the normalized m -fold primitive from -1 . Consequently, for $p \in \mathbb{N}$ and $n \geq 1$,

$$I_{-1}^n[t^p \text{up}(t)](x) = \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F((x+1)/2^{n+k}). \quad (7.10)$$

Proof. On the support, $\text{up}(x) = \mathcal{F}(x+1)$. Shift the global ladder of [theorem 3.1](#); for $m \geq 1$ the argument $(x+1)/2^m$ lies in $[0, 1]$, so the bounded F is sufficient. The polynomial formula follows exactly as in [theorem 4.1](#), now with base point -1 . $\square$

For fractional powers on the negative half of the support, it is natural to use $(-x)^\alpha$. If $-1 < x < 0$, substitution $u = x+1$ and a uniformly convergent binomial series give

$$\int_{-1}^x (-t)^\alpha \text{up}(t) \, dt = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} A_k(1+x). \quad (7.11)$$

At $x = 0$ the formula continues for $\Re \alpha > -1$ and then meromorphically. The even Mellin transform of the full bump is

$$\int_{-1}^1 |x|^\alpha \text{up}(x) \, dx = \frac{2D(\alpha+1)}{\alpha+1}. \quad (7.12)$$

8 Higher derivatives and mixed orders

8.1 Derivatives of monomially weighted Fabius functions

The corpus gives

$$F^{(j)}(x) = 2^{\binom{j+1}{2}} \mathcal{F}(2^j x) \quad (0 < x < 1). \quad (8.1)$$

Leibniz's rule immediately gives the derivative companion to the primitive formula.

Theorem 8.1 (Closed derivative formula). *For $n \in \mathbb{N}$, $\alpha \in \mathbb{C}$, and $0 < x < 1$,*

$$\boxed{\frac{d^n}{dx^n} (x^\alpha F(x)) = \sum_{j=0}^n \binom{n}{j} \alpha^{n-j} x^{\alpha-n+j} 2^{\binom{j+1}{2}} \mathcal{F}(2^j x).} \quad (8.2)$$

On $-1 < x < 1$, replace the final factor by $\mathcal{F}(2^j(x+1))$ to obtain the corresponding formula for $x^\alpha \text{up}(x)$ on a chosen branch of x^α .

Thus positive and negative operator orders are both finite on polynomial weights: positive orders use expanding dyadic arguments, while antiderivatives use contracting arguments. The signed extension is the common carrier that makes both directions globally meaningful.

8.2 A bilateral operator notation

For integer r define $\mathcal{D}^r$ by ordinary differentiation when $r \geq 0$ and by the normalized primitive I_0^{-r} when $r < 0$. For a monomial weight x^p , both signs of r produce finite sums. A useful research question is whether there is a natural fractional operator $\mathcal{D}^z$ satisfying a dyadic semigroup law and interpolating both (5.1) and (8.2). Classical Riemann–Liouville operators interpolate in the order of integration, whereas the present Newton series interpolates in the exponent of the monomial. The two complex parameters should be studied jointly.

9 Integrals and derivatives of the inverse Fabius function

9.1 The quantile area identity

Let $Q = F^{-1} : [0, 1] \rightarrow [0, 1]$. Strict monotonicity on the open unit interval and endpoint continuity are inherited from the corpus.

Theorem 9.1 (Inverse-power primitive). *For $0 < y \leq 1$, put $q = Q(y)$. Then for every $\alpha \in \mathbb{C}$,*

$$\boxed{\mathcal{Q}_\alpha(y) := \int_0^y Q(v)^\alpha dv = yq^\alpha - \alpha A_{\alpha-1}(q).} \quad (9.1)$$

Both sides are entire in α . At $y = 0$ the common continuous value is 0; this endpoint is understood by continuation rather than by the indeterminate symbol 0^α . At $y = 1$,

$$\int_0^1 Q(v)^\alpha dv = D(\alpha). \quad (9.2)$$

Proof. Substitute $v = F(x)$ in the Lebesgue–Stieltjes sense:

$$\int_0^y Q(v)^\alpha dv = \int_0^q x^\alpha dF(x).$$

Integration by parts gives $q^\alpha F(q) - \alpha \int_0^q x^{\alpha-1} F(x) dx$. The endpoint term at zero vanishes for every complex exponent because F is infinitely flat. Since $F(q) = y$, this is (9.1). The complete identity is the quantile representation of moments. $\square$

Corollary 9.2 (An elementary-looking inverse antiderivative). *An ordinary antiderivative of Q is*

$$\boxed{\int Q(y) dy = yQ(y) - F\left(\frac{Q(y)}{2}\right) + C.} \quad (9.3)$$

More generally, for $p \in \mathbb{N}_{>0}$,

$$\int_0^y Q(v)^p \, dv = yq^p - p \sum_{k=0}^{p-1} (-1)^k p - 1^k 2^{\binom{k+1}{2}} q^{p-1-k} F(q/2^{k+1}). \quad (9.4)$$

Differentiating (9.1) at $\alpha = 0$ gives

$$\int_0^y \log Q(v) \, dv = y \log q - A_{-1}(q). \quad (9.5)$$

Formula (9.3) is the exact area-under-an-inverse identity specialized to the Fabius primitive ladder. Its compactness is a useful diagnostic: differentiating the right side, the two Q' terms cancel because $\frac{1}{2}F'(Q/2) = F(Q) = y$.

For negative complete inverse powers,

$$D(-m) = 1 + mM(-m-1) = 1 + m \sum_{k \geq 0} \binom{m+k}{k} \frac{d_{k+1}}{k+1}. \quad (9.6)$$

Again the series has positive terms.

9.2 Weighted inverse integrals and the nonlinear hierarchy

Proposition 9.3 (Beta-weighted quantiles). *For $r \in \mathbb{N}$, $0 < y \leq 1$, $q = Q(y)$, and $\alpha \in \mathbb{C}$,*

$$\int_0^y v^r Q(v)^\alpha \, dv = \frac{y^{r+1} q^\alpha}{r+1} - \frac{\alpha}{r+1} \int_0^q x^{\alpha-1} F(x)^{r+1} \, dx. \quad (9.7)$$

Proof. After $v = F(x)$, write $F(x)^r \, dF(x) = d(F(x)^{r+1})/(r+1)$ and integrate by parts. $\square$

This introduces

$$H_{\beta,j}(q) = \int_0^q x^\beta F(x)^j \, dx, \quad (9.8)$$

a nonlinear hierarchy absent from the linear primitive ladder. Expanding the Cauchy kernel and using (9.7) yields:

Theorem 9.4 (Higher inverse primitive). *For $n \geq 1$,*

$$I_0^n[Q^\alpha](y) = \frac{y^n q^\alpha}{n!} - \frac{\alpha}{n!} \sum_{j=1}^n (-1)^{j-1} \binom{n}{j} y^{n-j} H_{\alpha-1,j}(q). \quad (9.9)$$

The linear Fabius ladder therefore solves the first inverse primitive completely, but higher inverse primitives expose powers of F . This is not a failure of the method: it identifies the exact new objects required at the next level.

9.3 Order-statistic interpretation

Let $X_1, \dots, X_n$ be independent copies of X .

Theorem 9.5 (Order statistics). *For real parts in the domain of the displayed expectation,*

$$I_0^n[Q^\alpha](1) = \frac{1}{n!} \mathbb{E} [\min(X_1, \dots, X_n)^\alpha]. \quad (9.10)$$

For $\Re \alpha > 0$ this is

$$I_0^n[Q^\alpha](1) = \frac{\alpha}{n!} \int_0^1 t^{\alpha-1} (1 - F(t))^n \, dt = \frac{\alpha}{n!} \int_0^1 t^{\alpha-1} \text{up}(t)^n \, dt. \quad (9.11)$$

In particular,

$$I_0^n Q(1) = \frac{1}{n!} \int_0^1 \text{up}(t)^n dt. \quad (9.12)$$

More generally, for $r, s \in \mathbb{N}$,

$$\int_0^1 y^r (1-y)^s Q(y)^\alpha dy = B(r+1, s+1) \mathbb{E} \left[X_{(r+1:r+s+1)}^\alpha \right]. \quad (9.13)$$

Proof. The minimum of n independent uniform variables has density $n(1-y)^{n-1}$; applying the quantile map Q gives the minimum of n independent Fabius variables. The survival formula for a positive power gives (9.11). The Beta density yields (9.13). $\square$

This probabilistic interpretation suggests studying the nonlinear integrals of F^j through order statistics rather than through direct functional equations.

9.4 Derivatives of the inverse

For $0 < y < 1$ and $q = Q(y)$,

$$Q^{(n)}(y) = \left[\left(\frac{1}{F'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{F'(x)} \right]_{x=q}. \quad (9.14)$$

Faà di Bruno's formula then gives

$$\frac{d^n}{dy^n} Q(y)^\alpha = \sum_{k=1}^n \alpha^k Q(y)^{\alpha-k} B_{n,k}(Q', Q'', \dots, Q^{(n-k+1)}), \quad (9.15)$$

where $B_{n,k}$ are partial Bell polynomials. Combining (9.14) with the Fabius derivative ladder expresses every derivative in terms of q , $\mathcal{F}(2^j q)$, and rational operations. Endpoint blow-up is encoded by the reciprocal factors $1/F'(q)$ and is naturally analyzed in the lower-Lambert coordinate already developed in the corpus.

10 Exact dyadic arithmetic, Thue–Morse sums, and q -binomials

10.1 Reciprocal-dyadic weighted primitives

Theorem 10.1 (Exact reciprocal-dyadic formula). *For $p, r \in \mathbb{N}$ and $n \geq 1$,*

$$I_0^n [x^p F(x)] (2^{-r}) = 2^{-r(n+p)-\binom{r}{2}} \sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} \frac{d_{r+n+k}}{(r+n+k)!}. \quad (10.1)$$

In particular the value is rational.

Proof. Insert $x = 2^{-r}$ into (4.2), then use (2.8) at index $r+n+k$. The power of two in each summand is

$$2^{-r(p-k)} 2^{\binom{n+k}{2} - \binom{r+n+k}{2}} = 2^{-r(n+p)-\binom{r}{2}},$$

independent of k . This cancellation is the arithmetic signature of the primitive ladder. $\square$

10.2 Exact inverse areas at reciprocal-dyadic ordinates

If $q = 2^{-r}$ and $y = F(q)$, then (9.4) simplifies similarly.

Theorem 10.2 (Dyadic quantile area). *For $p \in \mathbb{N}_{>0}$ and $r \in \mathbb{N}$,*

$$\int_0^{F(2^{-r})} Q(v)^p \, dv = 2^{-rp - \binom{r}{2}} \left[\frac{d_r}{r!} - p \sum_{k=0}^{p-1} (-1)^k p - 1^k \frac{d_{r+k+1}}{(r+k+1)!} \right]. \quad (10.2)$$

At $r = 2$, $q = 1/4$, $y = F(q) = 5/72$, the first six exact values are

p	$\int_0^{5/72} Q(v)^p \, dv$
1	1/72
2	1493/518400
3	53/86400
4	48781/365783040
5	10777/365783040
6	196806103/29847896064000

These were evaluated independently by exact rational arithmetic and checked by numerical quantile quadrature.

10.3 Arbitrary dyadic arguments and Thue–Morse signs

The corpus proves, for $0 \leq a \leq 2^N$,

$$F(a/2^N) = \frac{2^{-(\binom{N+1}{2})}}{N!} \sum_{h=0}^{a-1} (-1)^{s_2(h)} \sum_{j=0}^{\lfloor N/2 \rfloor} \binom{N}{2j} c_j (2a - 2h - 1)^{N-2j}. \quad (10.3)$$

Putting $x = a/2^r$ in (4.2) and applying (10.3) with $N = r + n + k$ gives an entirely finite closed form for every weighted primitive at every dyadic argument. It involves only rational centered moments c_j , binomial coefficients, powers of two, and the signed Thue–Morse sequence $(-1)^{s_2(h)}$. We leave it in this compositional form because expanding the three finite sums obscures the ladder structure and offers no computational advantage.

Write

$$(a; q)_N = \prod_{\ell=0}^{N-1} (1 - aq^\ell), \quad \left[\begin{matrix} N \\ j \end{matrix} \right]_q = \frac{(q; q)_N}{(q; q)_j (q; q)_{N-j}}, \quad (10.4)$$

with the usual empty-product convention [11]. The alternative translated half-base q -binomial identity in the corpus is, for $m, N \in \mathbb{N}$,

$$\mathcal{F}(m/2^N) = \frac{1}{2^{N^2} (1/2; 1/2)_N} \sum_{j=0}^N \frac{\left[\begin{matrix} N \\ j \end{matrix} \right]_{1/2}}{4^{\binom{j}{2}} (N+j)!} \sum_{h=0}^{m2^j-1} (-1)^{s_2(h)} (h - m2^j + R)^{N+j}, \quad (10.5)$$

independent of the translation parameter $R \in \mathbb{C}$. Substitution of (10.5) into each ladder term supplies a q -binomial version of the antiderivative formula. This explains how q -Pochhammer symbols enter the integral problem: not through integration itself, which is elementary in the ladder basis, but through exact evaluation of the contracted Fabius samples.

11 Numerical experiments and reproducibility

11.1 Independent representations

The accompanying program `fabius_integrals_experiments.py` uses two independent representations.

1. A dyadic grid with $2^{18} + 1 = 262145$ nodes is iterated under the integral/reflection fixed point for F . This checks pointwise finite primitives, shifted up-primitives, and inverse-quantile areas.
2. The exact rational recurrence (2.7) and a separate floating version through d_{900} check complete Newton sums, exact dyadic values, and negative-power tails without differentiating a flat function numerically.

The principal error maxima in the default run were

Test family	maximum absolute discrepancy
Finite F primitive versus direct quadrature	1.85×10^{-10}
Shifted up primitive versus direct quadrature	2.39×10^{-7}
Exact dyadic inverse areas versus quantile quadrature	4.04×10^{-9}
Grid value $F(1/4)$ versus $5/72$	3.23×10^{-12}

The larger up-error is caused by direct quadrature across a compact-support endpoint and by interpolation, not by instability of the closed form.

11.2 Fractional and negative complete integrals

With 900 moment terms the computed values include

α	$M(\alpha) = \int_0^1 x^\alpha F(x) \, dx$
-2	1.3148861440026451
-1	0.7582400004044057
-1/2	0.6073863107431697
1/2	0.4209681733402678
3/2	0.3146395582080128
$\sqrt{2}$	0.3218439940718418

Grid quadrature agrees to roughly 10^{-9} or better in these tests despite the negative powers.

11.3 Exact checks

For $p = 0, \dots, 12$, exact rational arithmetic verifies term by term that

$$\sum_{k=0}^p (-1)^k \binom{p}{k} \frac{d_{k+1}}{k+1} = \frac{1 - d_{p+1}}{p+1}. \quad (11.1)$$

The CSV files retain numerator/denominator strings, not floating reconstructions. The inverse checks at $q = 1/4$ likewise compute (10.2) with `fractions.Fraction` before comparison with quadrature.

12 Asymptotic tails for negative powers

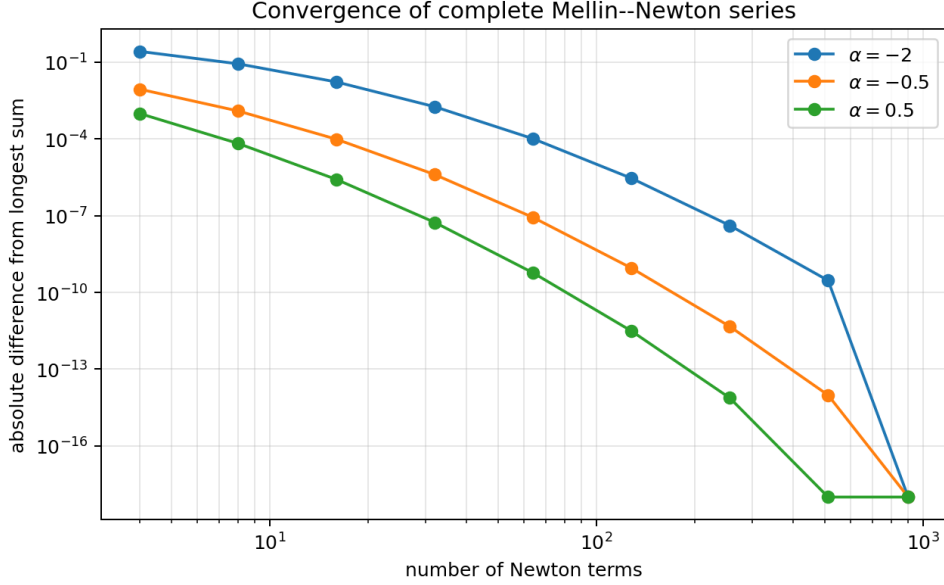


Figure 1: Convergence of the complete Mellin–Newton series. The reference is the longest available 900-term sum, so the final plotted error is clipped at machine scale rather than being a certified remainder.

12.1 Moment asymptotics inherited from the endpoint theory

Let $L = \log 2$. Combining the corpus’s effective dyadic endpoint theorem with (2.8) and Stirling’s formula yields the moment-scale structure

$$\log d_n = -\frac{(\log n)^2}{2L} + \frac{1}{2} \log n + C_M + \Psi(\log_2 n) + \mathcal{O}\left(\frac{(\log n)^2}{n}\right), \quad (12.1)$$

where Ψ is the same smooth nonconstant one-periodic correction whose Fourier coefficients are explicit Gamma–zeta values in the corpus, a standard Mellin-harmonic-sum phenomenon [9], and $C_M = C_F + \frac{1}{2} \log(2\pi)$. In the sharp source theorem the periodic argument is $\theta_n = \log_2 \lambda_n$, equivalently $\Psi(\lambda_n)$, where $\lambda_n - \log_2 \lambda_n = n$. Replacing θ_n by $\log_2 n$ changes the value by $\mathcal{O}((\log n)/n)$, already inside the displayed remainder. The exact theorem also retains the lower-Lambert saddle term $-\log^2 n/(2L^2 n)$ rather than absorbing it; the branch conventions are those standard for Lambert W [8]. Formula (12.1) is used here only to motivate a tail scale; none of the exact integral identities depends on it.

12.2 A first hazard approximation

For $a > 0$, the negative-power complete series has positive terms

$$t_k(a) = \binom{a+k-1}{k} \frac{d_{k+1}}{k+1}, \quad M(-a) = \sum_{k \geq 0} t_k(a). \quad (12.2)$$

The term envelope itself has logarithmic derivative

$$-\frac{d}{dx} \log t_x(a) \approx \frac{1}{x} \left(\frac{\log x}{L} - \left(a - \frac{3}{2}\right) - \frac{\Psi'(\log_2 x)}{L} \right). \quad (12.3)$$

For the tail, however, the correct endpoint-Laplace variable is $u = \log x$: $\int_N^\infty t_x(a) dx = \int_{\log N}^\infty e^u t_{e^u}(a) du$. The Jacobian contributes one additional unit to the power-law exponent. The effective endpoint denominator is therefore

$$H_a(N) = \frac{\log N}{L} - \left(a - \frac{1}{2}\right) - \frac{\Psi'(\log_2 N)}{L}, \quad (12.4)$$

which suggests

$$\sum_{k \geq N} t_k(a) \sim \frac{Nt_N(a)}{H_a(N)} = \frac{Nt_N(a)}{\frac{\log N}{L} - (a - \frac{1}{2}) - \Psi'(\log_2 N)/L}. \quad (12.5)$$

Ignoring the periodic derivative gives the simple estimator

$$\hat{R}_N(a) = \frac{Nt_N(a)}{\log N/L - (a - \frac{1}{2})}. \quad (12.6)$$

The experiment compares this with a 900-term reference tail. Representative ratios $\hat{R}_N/R_N$ are

a	$N = 50$	$N = 100$	$N = 200$	$N = 400$
1/2	0.950	0.974	0.990	1.000
1	0.959	0.980	0.994	1.002
2	0.987	0.996	1.003	1.010
3	1.037	1.022	1.019	1.021

At the smallest tails, double precision and the finite 900-term reference begin to matter; the agreement should therefore be read as structural evidence, not certification.

Conjecture 12.1 (Log-periodic negative-power tail). *For every compact interval $0 < a_0 \leq a \leq a_1 < \infty$, the remainder of (12.2) satisfies (12.5) uniformly as $N \rightarrow \infty$, with a relative error $o(1)$. A second-order expansion is obtained by the usual endpoint Laplace correction from the first two derivatives of the hazard.*

Missing proof step. One needs a sufficiently uniform differentiable version of the moment asymptotic along the integer ray, including control of the periodic derivative and second differences. The repository already formalizes the continuous negative-Laplace logarithm and its first periodic derivative estimate. The remaining bridge is a discrete transfer theorem from that saddle object to d_n and then to the weighted tail.

12.3 Acceleration

The Newton series converges faster than every algebraic rate but slower than a fixed geometric rate in the term index. Standard Richardson extrapolation on an arithmetic grid is therefore mismatched. The natural acceleration should use either

- the hazard variable $u_N = \log N/L - (a - 1/2)$, or
- a geometric sequence $N_j \approx N_0 2^j$, phase-locked so that $\log_2 N_j$ has constant fractional part and the periodic factor cancels.

Conjecture 12.2 (Phase-locked q -Richardson acceleration). *For fixed $a > 0$ and a fixed dyadic phase, there are explicitly computable linear combinations of the partial sums at $N, 2N, 4N, \dots, 2^r N$ that cancel the first r terms of the hazard expansion while preserving positivity-based enclosures after pairing successive transforms.*

This would connect the present exponent-Newton series with the corpus's existing q -Richardson methods, but the asymptotic variable is now the truncation index rather than a truncated sinc-product scale.

13 Further frontier directions

13.1 The nonlinear power hierarchy

The most immediate new object is $H_{\beta,j}$ from (9.8). Splitting at $1/2$, using $F'(x) = 2F(2x)$, and reflecting $F(1-x) = 1 - F(x)$ should produce coupled dyadic functional equations among $H_{\beta,j}$ and binomial combinations of lower powers. The following questions appear tractable.

Open problem 13.1. *Derive a triangular functional system for $H_{\beta,j}(x) = \int_0^x t^\beta F(t)^j dt$ under $x \mapsto 2x$ and $x \mapsto 1-x$, and determine whether integer β, j give exact rational values at dyadic x .*

Open problem 13.2. *Express the complete values $H_{\beta,j}(1)$ as moments of Fabius order statistics, then seek recurrences analogous to (2.7). The case $j = 2$ is the first genuinely nonlinear test.*

13.2 Two-parameter fractional calculus

The report interpolates the monomial exponent α while keeping the integration order n integral. One may instead define the Riemann–Liouville family

$$I_0^\nu[x^\alpha F(x)] = \frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} t^\alpha F(t) dt, \quad \Re \nu > 0. \quad (13.1)$$

Expanding as before gives

$$I_0^\nu[x^\alpha F(x)] = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} \frac{\Gamma(\nu+k)}{\Gamma(\nu)} x^{\alpha-k} J_{\nu+k}(x), \quad (13.2)$$

provided a meaningful analytic interpolation J_z of the discrete primitive ladder is chosen. The integer data $J_m(x) = 2^{\binom{m}{2}} F(x/2^m)$ suggests a q -difference or Mellin construction, but there is no canonical interpolation yet.

Conjecture 13.3 (Canonical interpolated ladder). *There exists a distinguished jointly holomorphic family $J_z(x)$ on a right half-plane, characterized by positivity on the real axis, suitable growth bounds, J_m at integers, and a functional relation interpolating $J'_{m+1} = J_m$. With this choice, (13.2) gives a jointly meromorphic two-parameter calculus in (α, ν) .*

13.3 Complex zeros and total positivity

The entire functions D and M satisfy

$$D(s+1) = 1 - (s+1)M(s). \quad (13.3)$$

The strict total positivity of $M(a+b)$ constrains its real zeros (there are none) and its Padé approximants. A natural complex-analytic program is to locate zeros of D and M , look for zero-free half-planes, and test whether the Stieltjes/Laplace structure yields a Laguerre–Pólya or Pick-type property after a suitable change of variables. Complete monotonicity alone does not force such a classification, so numerical claims should be made cautiously.

13.4 Characterization by polynomial closure

Open problem 13.4 (Rigidity). *Classify smooth functions f for which there are constants a_m, b_m such that every polynomially weighted primitive $\int P(x)f(x) dx$ is a finite linear combination of polynomials times $f(a_mx + b_m)$. Does nontrivial closure under all monomials force a dilation differential equation $f'(x) = cf(\lambda x)$, up to affine changes and elementary summands?*

The Fabius system should be a model case of a broader “finite primitive module” over the polynomial ring.

13.5 Certified computation

Positive negative-power terms make interval certification straightforward if exact rational moments are available. The main difficulty is a rigorous tail enclosure. Three routes are promising:

1. derive monotone upper and lower hazard bounds from the formal negative-Laplace estimates;
2. bound d_n directly by optimized beta integrals using effective endpoint inequalities;
3. perform ball-arithmetic evaluation of a moderate prefix and attach a rigorously controlled log-periodic saddle tail.

Such an algorithm would return certified enclosures for $M(-a)$, $D(-a)$, and inverse negative moments, not merely high-precision decimal approximations.

14 Lean formalization roadmap

The exact algebra is unusually formalization-friendly. A sensible dependency order is:

1. define J_m from the existing signed extension and prove $J'_{m+1} = J_m$;
2. identify J_m with the iterated interval integral on $[0, 1]$;
3. prove (4.1) by polynomial induction or finite telescoping;
4. specialize to monomials and derive the reciprocal-dyadic rational theorem;
5. establish theorem 3.2 using the probability measure already present in the development;
6. prove normal convergence of (5.1) on compact exponent sets;
7. define the entire functions D and M , prove (6.2), and formalize the forward-difference extraction (5.7);
8. define the inverse substitution as a pushforward/quantile integral and prove (9.1);
9. formalize the positive-half Mellin transform (7.7) and its one-pole continuation;
10. add order-statistic formulations after the nonlinear hierarchy has a stable API.

The only analytically substantial step is local normal convergence in α . It can be reduced to the already formalized flatness estimates plus a standard compact bound on $\binom{\alpha}{k}$. The integer polynomial theorem and inverse identity require no new deep analysis.

15 Conclusion

The antiderivative problem is governed by one exact mechanism. The signed differential law $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ turns into a contracting primitive ladder with triangular powers of two. Polynomial multiplication is then a nilpotent perturbation: repeated integration by parts terminates. Replacing an integer exponent by a complex exponent removes nilpotence but produces a normally convergent Newton series whose coefficients are precisely the dyadic ladder samples.

This viewpoint unifies several apparently separate topics. Complete weighted integrals are entire Mellin transforms of the Fabius distribution; their Newton coefficients are rational moments; those moments arise from the infinite sinc product through Bernoulli cumulants and Bell polynomials; exact dyadic evaluation imports Thue–Morse and half-base q -binomial formulae; and negative-power tails inherit the lower-Lambert log-periodic endpoint structure. The inverse function introduces no new object at first order: its primitive is the ordinary area-under-an-inverse formula closed by $F(Q/2)$. New nonlinear objects first appear at higher inverse order, where powers of F encode order statistics.

The most concrete next results are a rigorous log-periodic tail theorem, a dyadic functional system for $H_{\beta,j}$, and Lean formalization of the primitive ladder and exponent-Newton duality. These are narrow enough to attack directly and broad enough to connect the exact integral calculus back to the main arithmetic and asymptotic architecture of the repository.

A Technical convergence details

A.1 Rapid decay of the moments

For completeness, here is the elementary estimate used in [theorem 5.1](#). For every $A > 0$, endpoint flatness of up at 1 gives $\text{up}(t) \leq C_A(1-t)^A$ on $[0, 1]$. Then

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) dt \quad (\text{A.1})$$

$$\leq C_A n \int_0^1 t^{n-1} (1-t)^A dt = C_A n B(n, A+1) = \mathcal{O}_A(n^{-A}). \quad (\text{A.2})$$

No sharp endpoint asymptotic is needed.

A.2 Compact bounds for generalized binomial coefficients

For a compact $K \subset \mathbb{C}$, choose R with $|\alpha| \leq R$ on K . The recurrence $\binom{\alpha}{k+1} = \binom{\alpha}{k}(\alpha-k)/(k+1)$ and a finite initial adjustment imply

$$\sup_{\alpha \in K} \left| \binom{\alpha}{k} \right| \leq C_K (1+k)^{R+1}. \quad (\text{A.3})$$

Together with (2.9) and (5.2), this proves local uniform absolute convergence. Sharper asymptotics $\binom{\alpha}{k} \sim (-1)^k k^{-\alpha-1}/\Gamma(-\alpha)$ away from nonnegative integers are useful for tail analysis but unnecessary for existence.

A.3 Termwise exponent differentiation

Normal convergence permits differentiation with respect to α . For example,

$$\int_0^x t^\alpha (\log t)^m F(t) dt = \frac{\partial^m}{\partial \alpha^m} A_\alpha(x), \quad (\text{A.4})$$

with the derivative obtained termwise from (5.3). At $\alpha = 0$ this gives convergent series for antiderivatives of $(\log x)^m F(x)$; at $\alpha = -r$ it gives $x^{-r}(\log x)^m F(x)$. Thus logarithmic weights are included at essentially no extra cost.

B Selected exact formula table

Object	Exact representation
$I_0^n[x^p F]$	$\sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F(x/2^{n+k})$
$I_0^n[x^\alpha F]$	Same sum over $k \geq 0$ with $\binom{p}{k}$ replaced by $\binom{\alpha}{k}$; normally convergent.

Object	Exact representation
$M(\alpha)$	$(1 - D(\alpha + 1))/(\alpha + 1) = \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} d_{k+1}/(k + 1)$
$I_0^n[x^\alpha F](1)$	$\frac{1}{(n-1)!} \sum_{k \geq 0} (-1)^k \binom{\alpha}{k} d_{n+k}/(n+k)$
$I_0^n[x^p(1-F)]$	$p!x^{p+n}/(p+n)! - I_0^n[x^p F]$ on $[0, 1]$
$I_{-1}^n[x^p \text{up}]$	$\sum_{k=0}^p (-1)^k \binom{p}{k} n^{\bar{k}} 2^{\binom{n+k}{2}} x^{p-k} F((x+1)/2^{n+k})$
$\int_0^y Q(v)^\alpha dv$	$yQ(y)^\alpha - \alpha A_{\alpha-1}(Q(y))$
$\int Q(y) dy$	$yQ(y) - F(Q(y)/2) + C$
$I_0^n[Q^\alpha](1)$	$\frac{1}{n!} \mathbb{E}[\min(X_1, \dots, X_n)^\alpha]$

C Recursive corpus inventory

The following 76 .tex paths were returned by the pinned recursive documentation inventory. Historical copies are retained because novelty checking must include superseded formulae, not only the current exposition.

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1	papers/ubiq15/ubiq15-corrected.tex
2	papers/arXiv-2211.13570v1/document.tex
3	papers/arXiv-1702.05442v1/09-Function.tex
4	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
5	papers/arXiv-1502.06738v2/Lacunary_products.tex
6	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
7	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
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10	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
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16	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
17	archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex
18	archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
19	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
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21	archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
22	archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
23	archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
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29	archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
30	archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
31	archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
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34	archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
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36	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
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D Reproduction commands and generated files

From the extracted archive directory:

```
python fabius_integrals_experiments.py --output-dir .
latexmk -pdf -interaction=nonstopmode \
  Dyadic_Primitive_Ladders_Fabius_Rvachev.tex
```

The program generates:

- grid_inverse_power_checks.csv;
- fabius_finite_primitive_checks.csv;
- rvachev_shifted_primitive_checks.csv;
- complete_integer_exact_checks.csv;

- `fractional_complete_series_checks.csv`;
- `negative_power_tail_checks.csv`;
- `inverse_quantile_exact_checks.csv`;
- `newton_series_convergence.png`;
- `run_summary.txt`.

The script contains comments explaining every representation and every nontrivial scaling factor. NumPy and SciPy are used only for grid/quadrature checks; exact rational identities use Python's standard-library `Fraction` class.

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